

Writing the Proposal, Rigour and Dissemination

120 minutes · 6 self-check questions

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Assembling everything into a proposal that will be approved and funded, building in rigour rather than claiming it, and planning how findings will reach the people who can act on them.

Learning objectives

- Structure a research proposal to the standard expected by committees and funders
- Write a methods section detailed enough to be reproduced
- Build a realistic timeline and budget
- State limitations honestly without undermining the study
- Plan a data management strategy
- Design a dissemination plan that reaches participants and decision makers, not only journals

Proposal structure

- Title — specific, informative, usually naming design, population and setting
- Abstract — written last, covering background, aim, methods, expected outcomes; respect the word limit exactly
- Introduction and background — the problem, its magnitude, and the gap
- Literature review — the argument leading to your question
- Aim, objectives, research questions and hypotheses
- Methodology — design, setting, population, sampling, sample size, instruments, procedures, analysis plan
- Ethical considerations — approvals sought, consent, confidentiality, risk mitigation
- Timeline, budget, references, appendices with instruments and consent forms
- Reviewers commonly read the aim, then jump to sampling, sample size and analysis — those must survive isolated scrutiny

The methods section

- The test is reproducibility: could a competent stranger repeat this from what you wrote?
- Specify inclusion and exclusion criteria at the level of the individual participant
- State the sample size calculation with every input, not just the result

- Name the statistical tests against each objective, and say how missing data and confounding will be handled
- Pre-specify the analysis plan; deciding tests after seeing the data invites accusations of p-hacking
- For qualitative work, state the analytic approach, who will code, and how disagreements will be resolved
- Use the relevant reporting checklist while writing rather than afterwards: CONSORT for trials, STROBE for observational studies, COREQ or SRQR for qualitative work, PRISMA for reviews

IN PRACTICE

Reporting checklists are usually treated as a publication chore. Used at the proposal stage they function as a design checklist, and they surface the omissions reviewers would have found later.

Timeline, budget and feasibility

- Build the timeline backwards from the deadline, and add contingency — ethical approval alone commonly takes one to three months
- A Gantt chart makes dependencies visible: recruitment cannot begin before approval, analysis cannot begin before data cleaning
- Budget lines typically include personnel, transport, materials, participant compensation, data collection tools, software, transcription and translation, dissemination
- Transcription is routinely underestimated — an hour of interview takes roughly four to six hours to transcribe
- Justify every line; unexplained round numbers attract scrutiny
- State what you will do if recruitment is slower than planned — reviewers ask

Rigour and limitations

- Rigour is designed in, not asserted at the end
- Quantitative: adequate power, valid instruments, control of confounding, pre-specified analysis, appropriate handling of missing data
- Qualitative: audit trail, reflexivity, triangulation, thick description, transparent coding
- Limitations should name the threat, its likely direction, and what you did to reduce it
- "Time and financial constraints" is not a limitation; a convenience sample that may over-represent urban, literate respondents is
- Do not list limitations so severe they undermine the study — if one does, redesign rather than disclose and proceed

Data management and dissemination

- A data management plan states what data are collected, how stored, who has access, how backed up, how long retained and what is shared
- Follow the 3-2-1 rule: three copies, two media, one off-site
- Version your files by date and never overwrite raw data — all cleaning happens in a script or a documented copy
- FAIR principles — Findable, Accessible, Interoperable, Reusable — increasingly required by funders
- Dissemination should reach participants and their communities, not only journals: policy briefs, community feedback meetings, radio, professional bodies
- Feeding results back to the community that provided them is an ethical obligation, and it is what makes the next study possible
- Plan dissemination in the proposal, and budget for it

Worked examples

A limitations section that earns marks instead of losing them

A student writes: "Limitations included time and financial constraints, and a small sample size. More research is needed."

1. Time and money are constraints on the researcher, not limitations of the evidence — they tell the reader nothing.
2. "Small sample" is unquantified: small relative to what, and with what consequence?
3. Name the specific threat: the sample came from one urban clinic.
4. State the likely direction: urban patients are more literate and closer to care, so adherence is probably overestimated.
5. State the mitigation: findings are presented as applying to urban facility attenders, not to the district population.
6. Rewritten: "Recruitment from a single urban clinic likely over-represents literate, higher-income patients and probably overstates adherence; results should be read as applying to urban facility attenders."

The point: A good limitation names the threat, its direction, and what you did about it. Examiners reward the researcher who saw the problem first.

Building the timeline backwards from the deadline

A student has 12 months and plans: "Months 1-2 proposal, 3-8 data collection, 9-10 analysis, 11-12 writing." It is already unrealistic.

1. Ethical approval commonly takes 1-3 months and cannot start until the proposal is finished — so fieldwork cannot begin in month 3.
2. For 20 interviews at 1 hour each, transcription alone is 80-120 hours of work, which is weeks not days.
3. Analysis cannot begin until cleaning is done, and cleaning always takes longer than planned.
4. Supervisors need time to read: allow two weeks per full draft, and expect at least three drafts.
5. Working backwards from submission: writing 3 months, analysis 2, transcription 1, fieldwork 3, approval 2, proposal 1 — that is 12 with no contingency.
6. So either reduce scope now, or accept the plan will slip.

The point: Build the schedule backwards from the deadline and put the dependencies in a Gantt chart. Most overruns are visible at proposal stage to anyone who does the arithmetic.

Key terms

Gantt chart

A bar chart showing project activities against time, making dependencies and overlaps visible.

STROBE

The reporting checklist for observational epidemiological studies.

COREQ

The 32-item reporting checklist for qualitative interview and focus group research.

Data management plan

A document specifying how data will be collected, stored, protected, retained and shared.

FAIR principles

Findable, Accessible, Interoperable, Reusable — standards for research data stewardship.

p-hacking

Trying analyses until a significant result appears, then reporting only that one.

Policy brief

A short, non-technical summary of findings and recommendations aimed at decision makers.

Activity

Assemble a two-page proposal skeleton for your own study: title, aim, three objectives, design, sampling and sample size, analysis plan, and three limitations each stating direction and mitigation. Exchange with a partner and mark it against the relevant reporting checklist.

Further reading

- **EQUATOR Network — reporting guidelines** — Free index of every checklist; find yours before you write.
- **von Elm et al. (2007) — STROBE statement** — What an observational study must report.
- **Tong, Sainsbury & Craig (2007) — COREQ** — The qualitative reporting standard.
- **Wilkinson et al. (2016) — FAIR Guiding Principles** — Increasingly cited in funder data requirements.

Self-check questions

1. The best test of whether a methods section is adequate is:
 - a. It exceeds 1,000 words
 - b. A competent researcher could reproduce the study from it alone
 - c. It cites the most recent literature
 - d. It names the statistical software
2. Which is a properly stated limitation?
 - a. Time and financial constraints limited the study.
 - b. The convenience sample was drawn from one urban clinic and likely over-represents literate, higher-income patients, probably overstating treatment adherence.
 - c. More research is needed in this area.
 - d. The sample size was small.
3. Choosing statistical tests after inspecting the data and reporting only the significant ones is:
 - a. Exploratory analysis
 - b. p-hacking
 - c. Sensitivity analysis
 - d. Intention to treat
4. A proposal budgets three days to transcribe 20 interviews averaging one hour. This is:

- a. Generous
 - b. Unrealistic — transcription typically takes four to six hours per recorded hour
 - c. About right
 - d. Irrelevant to reviewers
5. The 3-2-1 backup rule means:
- a. Three researchers, two supervisors, one committee
 - b. Three copies, on two types of media, with one stored off-site
 - c. Three years of retention, two backups, one archive
 - d. Three drafts before submission
6. Feeding findings back to the community that provided the data is best understood as:
- a. Optional courtesy if funds remain
 - b. An ethical obligation that should be planned and budgeted from the start
 - c. A task for the funder
 - d. Necessary only for clinical trials
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Answers

1. **b.** Reproducibility is the standard. Length, citation currency and software names are incidental if a reader still could not repeat what you did.
 2. **b.** A useful limitation names the specific threat, its likely direction of bias, and lets the reader judge its effect. The others are generic statements that give the reader nothing to work with.
 3. **b.** Selecting analyses based on which produced significance inflates the false positive rate. Pre-specifying the analysis plan in the proposal is the protection against it; genuine exploratory work must be labelled as such.
 4. **b.** Twenty hours of audio implies roughly 80-120 hours of transcription. Underestimating this is one of the most common feasibility errors in qualitative proposals, and reviewers look for it.
 5. **b.** Three copies on two different media with one off-site protects against device failure, theft and fire — each of which has ended postgraduate projects that had only a laptop copy.
 6. **b.** Participants contributed time and risk on the understanding that knowledge would result. Returning findings respects that, and it is what sustains community willingness to take part in future research.
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